Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

Vitor Hugo de Oliveira Silva
NESPeD-LAB
Universidade Federal de Viçosa
Florestal, MG, Brazil
vitor.h.oliveira@ufv.br

Germano Barcelos dos Santos
NESPeD-LAB
Universidade Federal de Viçosa
Florestal, MG, Brazil
germano.santos@ufv.br

Fabrício Aguiar Silva
NESPeD-LAB
Universidade Federal de Viçosa
Florestal, MG, Brazil
fabricio.asilva@ufv.br

***Abstract*—Location-based social networks record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can cache content or reserve capacity ahead of demand. Two coarse questions about the next visit are usually enough: its category (the type of place) and its region (which part of the city). These are normally handled by separate models. We therefore test whether one model can learn both tasks, and what sharing a single model costs in per-task accuracy. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets (five U.S. states and one non-U.S. city, Istanbul), this improves next-category prediction over a standard place embedding at every dataset, by 0.2 to 6.3 points of macro-averaged F1, with every fold favoring it. We then train one multi-task model for both tasks and read both answers from a single saved model. On next region it outperforms the dedicated model at Texas and California, the two datasets with the most regions, by about one point, and at the other four it stays within a two-point margin registered before any result was read. On next category it outperforms the dedicated model at Florida, and the other five differences are equivalent to zero within half a point. One model therefore gives both answers in one forward pass at no cost beyond these bounds; a capacity control at California shows that a dedicated region model of the same size recovers the region gain there, so whether that gain comes from sharing or from model size remains open.**

***Index Terms*—mobility data; next-category prediction; next-region prediction; multi-task learning; check-in-level representation; location-based social networks**

I. Introduction

Location-based social networks, such as Gowalla [1], let people share the places that they visit as check-ins, which together record how people move through a city. Individual mobility is highly predictable in principle [2], so a service that anticipates the next move can prepare ahead of time, caching content where a user is heading [3] or provisioning capacity at the destination before demand arrives. Handover predictions already let cellular services adapt in advance at the network level [4]; at the city scale, check-in mobility analysis is likewise a design input for mobility-aware services [5].

Predicting the next place exactly, a single point of interest (POI), is hard and often more than a service needs. Two coarser questions are usually enough: the next category, the type of place such as food or shopping, and the next region, the part of the city that a user is heading to (a neighborhood-scale unit such as the U.S. census tract). The first captures intent, the second captures geography; we study whether one model should learn both at once.

Sharing one representation across tasks has a cost: in multi-task learning (MTL), one model does several jobs at once by sharing most of its parts, so the shared parameters can converge to a compromise optimal for neither task, helping one while hurting the other [6]. Our earlier work reported no consistent multi-task advantage for the paired category tasks and attributed it, in part, to this effect [7]; the useful question is where sharing helps, what it costs, and how to share so the gains hold and the cost stays small.

We propose two enhancements to that earlier model. First, we build a check-in-level representation: instead of one fixed vector per place, each check-in gets its own vector that holds its context (the time, nearby places, and recent visits). This adds a fourth level, the check-in, beneath the place, region, and city levels of hierarchical graph infomax [8], [9] (Fig. 1). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk (a cross-attention stack where the two tasks exchange semantic context) and a private spatial path for the region task (Fig. 2). We evaluate on two settings chosen to differ: five U.S. states of different sizes (Gowalla) and one non-U.S. city (Istanbul).[1]

Our contributions are as follows.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit, improving next-category prediction over a standard place embedding across all six datasets, by $+0.23$ to $+6.29$ points of macro-F1 (a balanced score that counts every category equally) (Table III). The direction is consistent even where the size is small, because a single fixed

[1]Code (model, representation, baselines, statistical tests): https://github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac. Both data sources are public: the category-annotated Gowalla dump (https://figshare.com/articles/dataset/gowalla_data/22126586) and Massive-STEPS [10].

vector per place cannot separate places that serve several purposes, which per-visit context recovers.

- A single model for joint next-category and next-region prediction, sharing semantic context across tasks while keeping a private spatial path for the region task; to our knowledge, the first work to treat fine-grained region as an end target of equal standing (Section II-B). In one forward pass, reading both answers from a single saved model, it outperforms a dedicated single-task region model at Texas and California, the two datasets with the largest region vocabularies, and at the other four it stays within a two-point margin registered before any result was read (statistical non-inferiority, assessed with two one-sided tests). At California a dedicated region model of the same size scores above it, so that gain is not attributed to sharing (Section VII). On category it outperforms the dedicated model at Florida, and the five remaining differences are equivalent to zero within half a point, even though the dedicated category model receives a per-dataset search (Table II).
- An empirical account, across five U.S. states and one non-U.S. city, of where the joint region gain appears. The two datasets where the joint model outperforms are the two with the largest region vocabularies, Texas and California, while the four with smaller vocabularies sit inside the two-point margin (Fig. 3). The grouping is reported as an observation rather than a law: the ordering does not hold inside the pair, since California has more regions than Texas and a slightly smaller gain, and region count co-varies with corpus size across these states. The joint and dedicated results of Table II average four random initializations; the representation comparison of Table III uses one (Section VI-B).

The rest of the paper presents related work (Section II), the problem (Section III), the method and setup (Sections IV and V), results (Section VI), and discussion and conclusions (Sections VII and VIII).

II. Background and Related Work

A. From place embeddings to per-visit embeddings

A place embedding transforms a POI into a vector so that similar or nearby places sit close together in the vector space. Deep Graph Infomax (DGI) [9], a general self-supervised method for graphs, learns such vectors on a place network linked by similarity, time, and distance, training them to tell the real network from a copy with shuffled node features. Hierarchical Graph Infomax (HGI) [8] adds geographic structure, organizing the network as place, then region, then city. Both produce one vector per place, so two visits to the same coffee shop look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [11], the closest example, learns a vector per visit by masking and reconstructing parts of a user's check-in sequence. Our representation, Check2HGI, also works at the check-in level but through a different construction. CTLE is a sequence model, a Transformer that reads the check-in sequence itself; Check2HGI remains a network model, keeping the hierarchical place-to-region-to-city network and the same infomax objective, now extended one level deeper, from individual places down to individual check-ins (Fig. 1). The order of a user's visits still enters the network, as links between consecutive check-ins (Section IV-A).

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE and a feature-concatenation control to separate the combination from contextualization alone and from feature injection.

B. Predicting the next category and the next region

A large body of work predicts the next place that a user will visit, the exact POI, with recurrent and attention models such as DeepMove [12], STAN [13], and GETNext [14]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. We choose them because they are what a mobility-aware service can act on, not to make the task easier; the region alone spans hundreds to several thousand classes. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, with a grid cell as the target [15]; our next-region task substitutes official neighborhood-scale units for grid cells. Our earlier work [7] paired static category classification with next-category prediction and found no consistent multi-task gain; this paper introduces the next-region task, adds the check-in-level representation, and reports, for each task, where the joint model outperforms a dedicated single-task model and where their difference stays within a stated bound (Section VI-B).

The field increasingly models several granularities at once; in those systems, category and region are auxiliary signals that help a primary next-place task (MCMG [16], HMT-GRN [17]). We study the pair as the object itself. To our knowledge, fine-grained region as an end target of equal standing, rather than an auxiliary coarse grid cell, is underexplored. The nearest exceptions do not study our exact pairing: DRRGNN [18] forecasts a person's next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition; a generative recommender [19] predicts category and region together, but only as auxiliary steps toward its next-place ranking.

C. Multi-task learning for mobility

MCARNN [20] jointly predicts activity and location, and a recent hierarchy-aware model continues the pattern [21]; in both, the location target is the exact place. The closest alternative to our setting is the cascade: predict the category first, then the place given the category [22]. CSLSL [23] is its current form, predicting in a chain (when, then what, then where), with location as the primary output and category as an intermediate step; CatDM [24] likewise uses the predicted category only to filter candidate places. We instead predict category and region in parallel, as end targets of equal standing in one model with a single forward pass, and we drop the next-place target entirely, so neither task is an intermediate step toward a third. On optimization, we are conservative by design: joint training with a fixed loss weighting is standard practice [6], not itself our contribution. Reports find that gradient-balancing methods, which re-weight the tasks' updates during training (PCGrad [25], Nash-MTL [26]), rarely improve on a well-tuned fixed weighting with two tasks [27]. We confirmed this during development, on an earlier preparation of the data, by screening nineteen loss and gradient balancers, including the two methods named above, at their default configurations and a single seed on two datasets, Alabama and Florida: none improved on fixed equal weighting on both tasks at both datasets. Two exceeded equal weighting on next-category at Alabama, Nash-MTL and scale normalization. At Florida, Nash-MTL fell below equal weighting on both tasks, and scale normalization stayed above it on next-category but collapsed on next-region. The reason is visible in the gradients. Measured during development on the same joint architecture, also on an earlier preparation of the data, the cosine similarity between the next-category and next-region updates on the shared trunk averages $+0.001$ across training (four seeds each on four Gowalla states: Alabama, Arizona and Florida, which are three of the five U.S. datasets reported here, and Georgia, which is not among the datasets this study reports; the largest per-dataset mean in absolute value is $+0.0032$). A balancer therefore has no conflict to resolve; this is a finding for this pair of tasks, not a general rule. (Istanbul, Alabama, Arizona, and Florida), is equivalent to zero at every one of them, with per-dataset means within two thousandths of zero, so the absence of conflict is not an artifact of this smaller set of runs.

III. Problem and Tasks

Given a user's time-ordered check-in history, we predict two properties of the next visit. The first is its *category*, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul's source collection maps its places onto the same seven labels. The second is its *region*, the place's census tract (for Istanbul, the *mahalle*, a municipal neighborhood). We do not predict the exact next place; both properties are easier to learn and, for most uses, enough. Region, unlike category, is a large label set: we frame next-region as classification over candidate regions, from 520 classes (Istanbul) to 8,501 (California; Table I).

The motivation is practical: the category says what type of place comes next, hence what a user will want; the region says where, hence where to prepare. The mobility literature supports such preparation: city services have long been built on location-based social network traces [28]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate, and ties this structure to infrastructure planning and adaptive strategies [5]. That analysis reads past visits; we predict two properties of the next one, and aggregated over users, those predictions point to where demand is heading before it arrives. A census tract is a neighborhood, not a radio cell. We therefore scope our claims to neighborhood-level preparation: demand and load anticipation, caching content ahead of time, and capacity planning. Cell association and handover are radio-level decisions and remain out of scope. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. Method

A. The check-in-level representation

We want each visit, not each place, to have its own vector; we call the resulting representation Check2HGI (Section II-A). Fig. 1 shows the full data flow, from check-ins to the two outputs.

We build a graph with four levels: the check-in, its place, the place's region (a census tract), and the city. Edges connect each level to the one above it and link a user's consecutive check-ins with a weight that decays as the time gap between the visits grows; same-place visits connect through their shared place node one level up, and nearby places are linked at the place level. Each visit's category, time of day, and day of week enter as input features of its node, not as edges. Four elapsed-time features join them: the time since the user's previous visit and since the user's first visit, both on a logarithmic scale, the gap to the previous visit within the same day, and an indicator for a user's first visit. Each is measured up to the visit itself, so a node describes the visit and the history preceding it, never anything that follows. These give the representation a sense of tempo, distinguishing a visit that follows minutes after the last one from a visit that opens a new outing, which a categorical hour-and-weekday encoding alone cannot express. The consecutive-visit edges run in one direction only, from an earlier visit to a later one, for the same reason: a target is predicted from a user's past, so the representation is built from the past alone. This keeps a hierarchical graph's place-to-region-to-city geography and adds the check-in as a bottom level. We train the graph mainly with an infomax objective, under which each vector learns to match its real neighborhood and reject a shuffled one [8], [9]. Two small label-free auxiliary terms are added (weights 0.3 and 0.1): a masked reconstruction of each place's aggregated category features, and an anchor to a place embedding pre-trained, label-free, on the same data. The training uses no task label: it never sees the next category or the next region. From the trained graph, we extract one 64-dimensional vector per check-in (its region nodes have trained vectors as well), so a model sees a sequence of per-visit vectors rather than repeated per-place ones.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts the next visit's category and region at once. Fig. 2 shows the design. Each task has its own input. The category task reads the window of per-visit vectors (the semantic stream); the region task reads the same window of visits, each visit now represented by the trained vector of its region node from the same graph (the spatial stream). Both pass through private per-task encoders (a small input network

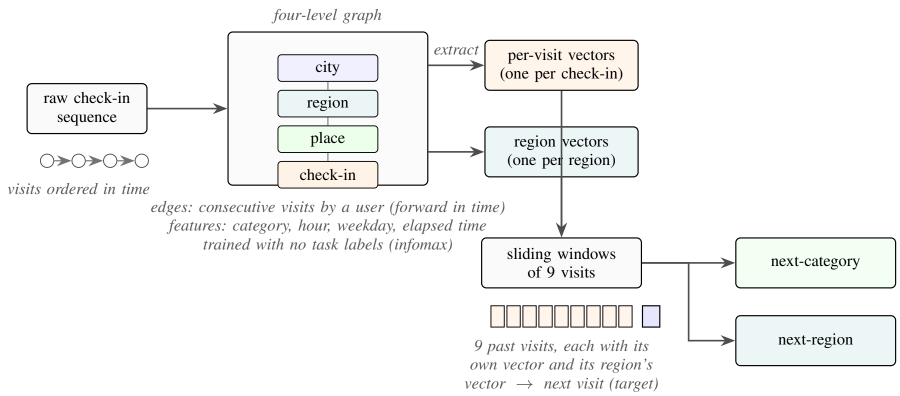

Fig. 1. From a check-in sequence to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) trained without task labels. The graph yields two sets of vectors, one per visit and one per region; sliding nine-visit windows of each feed a single model that outputs the next category and the next region.

per task, with no shared weights) into the shared trunk, a cross-attention stack of two blocks, so each prediction still uses both inputs. In each block, attention lets each stream read the other's features, while each keeps its own feed-forward weights. The tasks therefore share by exchanging information between per-task streams, not by owning hidden layers in common. The trunk then feeds a category output and a region output; the region output also keeps a private spatial path: a small branch inside the one model (not a second model) that reads the spatial input window, bypassing the shared trunk. The category task does not touch this branch. Together, the region task's encoder, its output, and this private path form the region pathway.

The training loss is a fixed-weight sum of the two outputs,

$$L = 0.5\, L_{\text{cat}} + 0.5\, L_{\text{reg}}, \tag{1}$$

where L_{reg} is a plain unweighted cross-entropy loss and L_{cat} is a cross-entropy loss with logit adjustment [29]: $\tau \log P_{\text{train}}(y)$, with $\tau = 0.5$, is added to the category logits during training only, which shifts the decision boundary toward the balanced posterior that macro-F1 rewards. Inference uses the unadjusted logits. The weights and τ were selected once on validation during development and held fixed across all six datasets. Class weighting, tested on both outputs, lowered both region accuracy and category macro-F1, and logit adjustment replaces it on the category output. The two choices differ by task because the two metrics do: macro-F1 weights every category equally, which the adjustment targets directly, whereas Acc@10 rewards ranking the frequent regions well, so the region output is left unadjusted. This is standard fixed-weight joint training [6], kept simple by design so that any improvement over the dedicated single-task models comes from the shared representation, not from an adaptive weighting scheme. The dedicated category model receives the same adjustment, so the comparison between them is not affected by it.

The joint model includes the shared trunk and both task outputs, so it is larger than either dedicated model, and a forward pass costs more compute than running the two small dedicated models: the joint model has about 4.2 million parameters at Alabama against 1.9 million for the two dedicated models combined (5.2 against 2.8 at California). What the single model provides is operational rather than arithmetic: one artifact to train, version, and deploy, and one forward pass whose one set of inputs produces both answers at once.

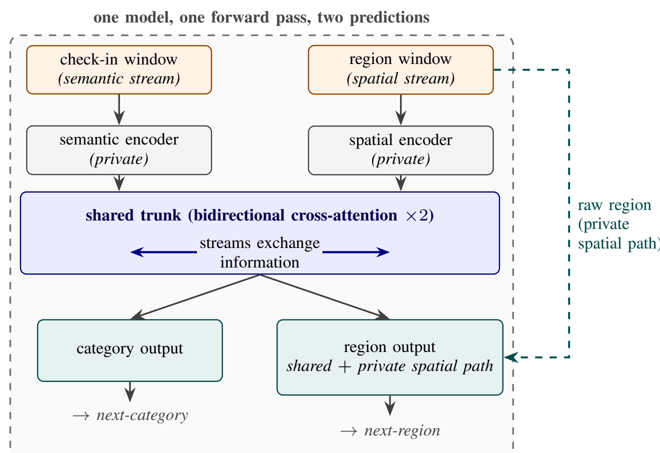

Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into the shared trunk, the only place where the two tasks interact. The category output uses the trunk's output alone; the region output combines it with a private spatial path. One model, one forward pass, two predictions.

V. Experimental Setup

This section describes the datasets, data splits, evaluation metrics, statistical tests, and comparison models used in the experiments.

A. Data

Table I summarizes the six datasets. Five datasets come from Gowalla, a location-based social network, and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [1]. The sixth dataset contains check-ins from Istanbul in the Massive-STEPS collection [10]. We use Istanbul to examine the findings outside the United States. The U.S. datasets range from about 114,000 check-ins and 1,109 regions in Alabama to about 3.2 million check-ins and 8,501 regions in California.

All six datasets use the seven place categories defined in Section III. The source labels for Istanbul are mapped to these categories. A region is a census tract in the U.S. datasets and a mahalle in Istanbul.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits, we order the visits by time. We then form overlapping sliding windows of nine visits and use the next visit as the target. A new window starts at each visit, which provides more examples for both tasks. Near the end of a user's history, several start positions can produce shorter, padded windows with the same final visit as the target. We remove these duplicates and keep only the full-length window that ends at this target. The prediction horizon is usually short, but some time gaps are much longer. The median time from the last visit in a window to its target ranges from 0.4 hours in Florida to 5.5 hours in Istanbul. Across the datasets, 5 to 27 percent of the targets occur more than 3 days later.

Splitting. We use stratified five-fold cross-validation and split the data by user. All windows from one user remain in the same fold. Therefore, the overlapping windows cannot cross the training and test sets, and the visits of a test user never appear in training. The held-out fold provides the validation data, and we do not reserve a third split. We stratify the split by the next-category label because the seven classes are imbalanced. Food is the majority class in every dataset, ranging from about 25 percent of visits in Florida to 34 percent in Alabama (Table I).

Integrity of the representation. Before using Check2HGI in the prediction models, we need to check whether it carries information from the validation data. The prediction models are trained without the validation users, but Check2HGI is trained once on the whole dataset and has seen their visits. Its training objective does not use the next-category or next-region labels. We therefore check whether whole-dataset training changes the results.

Configuration search. The search was not uniform across the three model families, so its scope is stated here. For the dedicated category model, batch size was searched at all six datasets, over five folds at Istanbul, Alabama and Arizona and on a single fold at Texas, and the learning rate at four of them, over five folds at Istanbul, Alabama and Arizona and on single folds at Texas; Florida and California carry the large-dataset value, which only the single-fold Texas screen tested. The selected learning rate differs by dataset. For the joint model, batch size and the category-head learning rate were searched at Istanbul, Alabama, and Arizona over five folds and screened at Florida; Texas and California carry the configuration transferred from those searches. The dedicated region model uses one configuration throughout, with only the logit-adjustment strength tested and set to zero for that task. Section VII returns to what the uneven joint search may mean for Texas and California.

Whole-dataset training. The question is whether using all users to train Check2HGI gives the prediction models an advantage. To check this, we built a new representation for each fold using only its training users. This check covers three datasets at one seed, on an earlier build of the representation. The differences were -0.33 to $+0.01$ Acc@10 points for region and 0.00 to $+0.29$ macro-F1 points for category.

However, the region comparison uses region vectors directly. For category, a graph built only with training users has no visit vectors for validation users. We therefore used one vector per place and kept only windows whose input places occurred in training. These windows cover 67 to 87 percent of the validation data. The comparison does not cover information specific to each visit or places unseen in training. Within this coverage, whole-dataset training did not meaningfully change the results.

C. Metrics and statistical tests

For the category task, we report macro-averaged F1 (macro-F1). It is the mean of the F1 scores for the seven categories, so each category has equal weight. Its reference point is the majority-class floor, obtained by always predicting the most common category. For the region task, we report accuracy at ten (Acc@10). This measure is the fraction of test visits for which the true region is among the model's ten highest-scoring predictions. If the true region does not occur in the training data for that fold, the visit counts as an error. The reference point for this task is the dedicated model.

A claimed gain and a claimed match require different tests. A non-significant difference does not provide evidence of a match. A written analysis plan, fixed during development and before any result was read, assigned a superiority test to next-category prediction and a non-inferiority test to next-region prediction. These assignments apply to each task across the datasets. The plan did not define a superiority test for next-region prediction. Therefore, the two next-region gains in Section VI-B are secondary results outside the plan. On next category the plan registered no equivalence margin, so a difference that fails the superiority test is reported as unresolved rather than as a match.

The plan registered a paired Wilcoxon signed-rank test on the matched fold differences. The final evaluation uses four seeds, each of which determines a different user partition and repeats the five-fold experiment. This design produces $4 \times 5 = 20$ fitted models per configuration. The reported analysis aggregates the five folds within each seed and reports a paired t-test on the four per-seed means (n=4), together with the 90% confidence interval for the paired difference. At this seed-level footing, the exact one-sided Wilcoxon p-value cannot be less than 0.0625. The code release includes both tests and documents this departure from the plan.

A *seed* is one complete repetition of the five-fold experiment. It determines the random initialization and the user partition, so each seed produces a different division of the users. Within one seed, the compared models use the same partition. The paired t and TOST analyses compare the four per-seed means. We apply a Holm correction [30] across the six next-category comparisons and, separately, across the six next-region comparisons.

The non-inferiority claim states that the joint model is no worse than the dedicated model by more than a two-point margin. We test this claim with the two one-sided tests (TOST) procedure [31]. The analysis plan also fixed the two-point margin in advance. A mobility-aware service acts on which region will be busy, rather than on a single position in the ranking (Section III). A two-point change in Acc@10 is below the level at which this service would behave differently. Across the four user partitions, the standard deviation of the paired difference ranges from 0.02 to 0.16 points. The intervals at Istanbul, Arizona, and Florida are narrow enough to support a margin as small as one point; Alabama's is not, and among these four datasets it has the largest region difference (Section VI-B).

D. Baselines

We use the comparison models for two purposes: to compare each task with established methods and to evaluate the representation.

Task comparisons. For next-category prediction, we use POI-RGNN [32], which we re-implement from its published architecture and hyperparameters. We also use a Markov model that predicts the category that most often follows the recent categories. We select the best Markov order for each dataset. For region prediction, we compute a Markov-1 floor over region transitions. It uses the same sliding windows and fold splits as our models.

The primary next-region comparison is HMT-GRN [17], an end-to-end recurrent model trained as one unit from the raw check-ins. We evaluate it on the same data, folds, and random initializations. We keep its shared multi-task structure and add a region-transition prior that is built from the training data in each fold. A version built from the whole dataset inflated region accuracy by 13 to 27 points. Only the HMT-GRN comparison model uses this prior; our joint and dedicated models do not. We remove the graph components and hierarchical beam search because they support exact next-place prediction, which we do not study. The resulting model predicts region as one of its original targets, but it is not a reproduction of the complete published system.

We also run STAN [13]. We re-implement its published architecture and train it from the raw check-ins with its own representations and sequence construction under one fixed configuration. We adapt its output to rank candidate regions, which are spatial objects compatible with its candidate-distance matching. We do not adapt STAN to category prediction because distance matching does not apply to categories. We report ReHDM [33] as a reference under its published protocol.

Representation controls. Two controls test whether the category gain comes from our representation design rather than from contextualization in general or from additional features. CTLE [11] is the closest prior contextual check-in representation. We train it end to end at Florida. At Alabama, Arizona, and Istanbul, we keep its weights fixed and do not fine-tune them. The learned representations follow their own training scopes: HGI is pre-trained once on the whole dataset, as our representation is, while CTLE is trained for each fold using training users only. The second control adds raw features from each visit to a standard place embedding and uses the same prediction model.

A final control uses the standard place-level HGI representation [8] to measure what the per-visit representation adds. All other choices, including the nine-visit window, were fixed during development. The released code provides the full training configuration for every model (footnote 1).

VI. Results

A. The check-in-level representation improves the category task

The check-in-level representation reaches a higher next-category macro-F1 than a standard place embedding (HGI [8]) at every dataset (Table III). The comparison is controlled: target, single-task model, training configuration, windowing, epoch budget, logit adjustment, and folds are identical; only the input representation changes (a vector per visit versus a vector per place), so the difference isolates the representation. The check-in-level column is the same dedicated single-task category model reported in Table II, read at seed 0 so that the two columns share one fold partition; the place-level column is the same model with its input swapped. The difference ranges from $+0.23$ macro-F1 at Florida to $+6.29$ at Istanbul, and all five folds favor the check-in-level representation at every one of the six datasets. A paired test over the five matched folds separates the two representations at every dataset except Florida, where the direction is unanimous across folds but the difference does not reach significance ($p = 0.07$). What the comparison establishes is therefore a consistent direction rather than a large effect: the input representation is one condition that affects performance here, and it acts the same way everywhere it was measured.

The geometry of the vectors indicates why the representation helps this task in particular: the silhouette score by category of the representation's vectors (how tight and well separated the seven labeled groups are, on a -1 to 1 scale) is about 0.57 against about 0.00 for the place embedding, and their nearest-neighbor category purity (the share of nearest neighbors with the vector's own category) is about 0.98 against about 0.78 (both averaged over the five U.S. states). Both measures were taken on an earlier build of the representation, with the vectors aggregated per place, so they characterize the representation family rather than the exact configuration evaluated in Table II. The same geometry does not separate regions, so the benefit is category-only (the joint model's spatial stream instead reads the region-level vectors of the same graph, Section IV-B). CTLE [11], the closest prior contextual embedding, also gives each visit its own vector, but it pretrains on place identifiers and timestamps alone, so the category vocabulary never enters its training, whereas our graph reads each visit's category and time of day as input

TABLE I

DATASET STATISTICS, ORDERED BY REGION COUNT: FIVE GOWALLA STATES AND ISTANBUL (MASSIVE-STEPS). ALL SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS: SLIDING NINE-VISIT INPUTS PLUS THE NEXT-VISIT TARGET; LENGTHS: PER-USER CHECK-IN COUNTS; DENSITY: $100 \times$ CHECK-INS/(USERS $\times$ POIS); MAJORITY: SHARE OF NEXT-VISIT LABELS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)	Majority (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

TABLE II

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1; REGION IS ACC@10. BOLD AND $\uparrow$ MARK AN IMPROVEMENT OVER THE DEDICATED MODEL WITH PAIRED STATISTICAL SUPPORT (HOLM-CORRECTED WITHIN EACH TASK; SECTION VI-B). IN THE REGION COLUMNS $\approx$ MARKS A DIFFERENCE THAT STAYS WITHIN THE TWO-POINT MARGIN REGISTERED IN ADVANCE (TWO ONE-SIDED TESTS); ALL FOUR ARE SMALL DEFICITS. CATEGORY ENTRIES CARRY NO SUCH MARK: NO MARGIN WAS REGISTERED FOR THAT AXIS, AND THE FIVE UNMARKED DIFFERENCES ARE BOUNDED BY THE INTERVALS THEMSELVES (SECTION VI-B).

		Next-category (macro-F1)				Next-region (Acc@10)				
Dataset	Regions	Markov-K	POI-RGNN	Dedicated	Joint (ours)	HMT-GRN	ReHDM	STAN	Dedicated	Joint (ours)
Istanbul	520	24.55	30.12	35.34 $\pm$0.01	35.42 $\pm$0.06	60.4	69.33	61.86	75.16 $\pm$0.01	75.08 $\pm$0.05$^{\approx}$
AL	1,109	20.50	23.80	30.77 $\pm$0.07	30.59 $\pm$0.07	57.05	65.38	60.72	70.12 $\pm$0.10	69.24 $\pm$0.16$^{\approx}$
AZ	1,547	23.92	27.64	34.57 $\pm$0.04	34.57 $\pm$0.06	43.70	53.00	49.86	59.48 $\pm$0.07	59.04 $\pm$0.22$^{\approx}$
FL	4,703	29.74	34.49	37.35 $\pm$0.02	**37.55** $\pm$0.07$^{\uparrow}$	63.74	64.49	72.99	76.69 $\pm$0.01	76.54 $\pm$0.01$^{\approx}$
TX	6,553	28.67	33.03	36.33 $\pm$0.01	36.19 $\pm$0.04	53.85	48.81‡	61.67†	64.94 $\pm$0.01	**66.15** $\pm$0.07$^{\uparrow}$
CA	8,501	27.58	31.78	35.63 $\pm$0.01	35.63 $\pm$0.02	49.61	50.26‡	58.52†	63.48 $\pm$0.03	**64.54** $\pm$0.04$^{\uparrow}$

HMT-GRN (region-native, on our folds, scored on visits whose region appears in training, >99% of test visits in every dataset and fold) is the primary external comparison; STAN (our re-implementation) and ReHDM (its own protocol) are references. Markov-K: strongest order per dataset. †STAN partial folds: TX 4/5, CA 2/5 (seed 0). ‡ReHDM at TX and CA: a single seed. Joint and dedicated entries: four seeds $\times$ five folds; $\pm$: sd across seeds. For the dedicated category model, batch size was searched at every dataset and the learning rate at four of the six; the joint model was searched at the four smaller datasets, and TX and CA carry the joint configuration transferred from those searches. The dedicated region models use one fixed configuration throughout.

features. At Florida, we fine-tune CTLE together with the task model, using its authors' defaults and the same 64 dimensions and windowing as ours. It reaches 33.45 macro-F1 at its best epoch and 29.69 at its final one (an epoch is one pass over the training data), below the place embedding under the same best-epoch rule. With frozen weights, fed to the same single-task model, CTLE repeats the ordering at Alabama, Arizona, and Istanbul. These CTLE results establish the ordering between the two representation families rather than a difference against the values in Table III. A feature-concatenation control joins the place embedding with the same raw per-visit features that our graph reads (the category one-hot plus hour and day-of-week encodings) and feeds the result to the same single-task model. On the category axis, at Alabama, Arizona, and Florida this control raises the place embedding by $+2.0$, $+1.7$, and $+0.8$ macro-F1, against place-to-check-in gaps of $+1.62$, $+2.58$, and $+0.23$ at the same states (Table III). Most of the category difference is therefore already available in the raw per-visit features; what the check-in-level representation adds beyond them on this axis is small, and this control does not separate it.

B. One model, two tasks

One model serves both tasks, and every difference on both axes stays inside the bound its axis supports. On next region it outperforms the dedicated model at Texas and California, the two datasets with the most regions, and at the other four it stays within the two-point margin registered before any result was read (statistical non-inferiority, assessed with two one-sided tests). On next category it outperforms the dedicated model at Florida, and at every dataset the difference between the two is equivalent to zero within half a point, simultaneously across all six.

Half a point is worth placing against the two comparisons that give the category axis its scale. The input representation moves the same metric by $+0.23$ to $+6.29$ points, up to thirty-two times the largest difference the choice of architecture makes, and the joint model stands at least 3.06 points above the strongest external baseline at every dataset. Whether one model or two produce the category prediction is therefore the smallest of the three effects at work here, which is the sense in which one model can replace two on this task. For scale: always predicting the most common category reaches between 5.7 and 7.3 macro-F1 depending on the dataset (the majority-class floor, lowest at Florida, whose category mix is the most even), and a random region top-ten guess is right at most about two percent of the time (Section V-C).

Table II reports the single joint model against the dedicated single-task ceilings (the level a joint model is usually expected, at best, to reach). Batch size was searched for the dedicated category model at every dataset and the learning rate at four of the six, with the joint model searched at the smaller datasets; Section V-B states the coverage per knob and dataset. Texas and California carry the joint configuration transferred from those searches, and the dedicated region models use one fixed configuration throughout. Throughout, every reported model is one saved artifact per fold, read at its validation-selected epoch: each dedicated model at its task's best epoch, and the joint model at the epoch selected by its joint validation score (the geometric mean of the two task metrics), with both tasks read from that one model. This convention is the one a deployed system can actually serve, since it commits to a single saved model per fold. The alternative, reading each task at its own best epoch, describes no single saved model: it reports two different saved models as though one system produced both. That alternative is more favorable to the joint model, by at most 0.23 macro-F1 and 0.93 Acc@10 points as the largest gap at any one seed, and it is favorable enough to change the verdicts, turning four further category comparisons and two further region comparisons into improvements that survive the same Holm correction applied here. The stricter convention is reported here for that reason, and every verdict in this paper is the one it yields.

Fig. 3 plots the signed differences ordered by region count. On category, every difference is small in both directions, from -0.19 macro-F1 at Alabama to $+0.19$ at Florida, and Florida is the only one the tests resolve. On region (Acc@10), the joint model outperforms the dedicated model at Texas ($+1.21$) and California ($+1.06$), and at the other four datasets it stays within the two-point margin. The two datasets where it outperforms are the two with the largest region vocabularies, which is where the region task is hardest and where the dedicated model has the most to gain from an auxiliary signal. The ordering is not monotone inside that pair, since California has more regions than Texas and a slightly smaller gain, and region count co-varies with corpus size across these states, so the grouping is read as an observation about where the benefit appears rather than as a law relating gain to region count. Why the benefit appears there is a separate question: at California, a dedicated region model widened to the joint model's size scores above the joint model (Section VII), so the gain at that dataset is not attributed to sharing between the tasks, and Texas has no control of matched size.

The tests behind those verdicts are reported next, each entry giving the point estimate and its 90% confidence interval. On category, Florida is the one dataset where the difference survives the Holm correction across the six ($+0.19$; $+0.14$ to $+0.25$; corrected $p = 0.011$), with 19 of the 20 folds favoring the joint model. The other five differences do not survive it, and because the equivalence margin was registered for the region axis only, they are reported by the bound their intervals support rather than as matches: Istanbul ($+0.08$; $+0.01$ to $+0.15$), Arizona (-0.00; -0.04 to $+0.03$), California (-0.00; -0.03 to $+0.02$), Texas (-0.13; -0.19 to -0.08) and Alabama (-0.19; -0.33 to -0.04). The intervals themselves carry what can be said about magnitude: the widest of them reaches 0.34 points from zero, at Alabama, which bounds all six differences within half a point of zero at once. The bound is read off the intervals rather than established by a further test.

On region, Texas ($+1.21$; $+1.13$ to $+1.29$; corrected $p = 0.00013$) and California ($+1.06$; $+1.03$ to $+1.08$; corrected $p < 10^{-4}$) survive the correction, with all 20 folds favoring the joint model at both. The remaining four stay within the registered two-point margin, and their direction is stated rather than rounded away: all four are deficits, at Alabama (-0.87; -1.00 to -0.75), Arizona (-0.44; -0.62 to -0.25), Florida (-0.16; -0.19 to -0.13) and Istanbul (-0.08; -0.16 to -0.002), and every one of the four intervals lies entirely below zero. A test in the reverse direction, applied post hoc to the same six region comparisons and corrected across them, resolves three of the four; Istanbul's is not resolved, its interval reaching to within two thousandths of zero. Each of the four stays inside the margin, which is what the pre-registered analysis asked of them, but none of them is a tie.

The joint model is also above every external baseline reported, on both tasks, across all six datasets (Table II): on region, the primary region-native comparison HMT-GRN [17], STAN [13] trained from the raw check-ins, and a ReHDM reference [33]; on category, POI-RGNN [32] and the Markov-K floor. These externals run on their own embeddings, so this comparison also includes the representation advantage of Section VI-A; the controlled comparison for the joint model remains the Dedicated column of Table II, which uses the same representation, windows, and folds. The first-order Markov region floor of Section V-D, computed under our windows and folds, reaches 51 to 72 Acc@10 across the datasets; the joint model exceeds it by 4.1 to 10.0 points on all six datasets.

That floor is also above the three external region systems at most datasets, and the comparison deserves a direct word rather than being left for the reader to assemble. HMT-GRN falls below it at all six, the ReHDM reference at three, and STAN at four. Two facts about how these numbers were produced bear on that comparison. The first concerns the floor: it is computed under our own sliding-window protocol and fold splits. Those windows advance one visit at a time, so the region of the last visit is a strong predictor of the next one, and a first-order transition table reads exactly that signal. At Alabama the target region is the last visited region in 32.9 percent of windows. The second fact concerns the three systems, which do not meet the floor on equal terms. HMT-GRN is evaluated on the same data, folds, and initialization as our models. STAN runs on the same folds, but builds its own embeddings and its own sequences from the raw check-ins. The ReHDM reference runs under its own published protocol, so its result is not measured on our windows or folds at all.

Neither fact establishes why the floor lies above the three systems, and we do not claim a single explanation. We treat the floor, not the external systems, as the reference the region task has to clear, and we read the external columns as a comparison against published designs rather than as the standard for this task. The dedicated and joint models are above the floor at every dataset.

C. External validity (Istanbul)

The joint-model results hold for a non-U.S. city under the same representation, sliding-window protocol, and training setup as the U.S. states. At Istanbul, both differences are within

TABLE III

NEXT-CATEGORY MACRO-F1: THE CHECK-IN-LEVEL REPRESENTATION AGAINST A STANDARD PLACE EMBEDDING (HGI). SAME SINGLE-TASK MODEL, TRAINING CONFIGURATION, FOLDS, SLIDING WINDOWS, EPOCH BUDGET, AND LOGIT ADJUSTMENT IN BOTH MODELS; ONLY THE INPUT REPRESENTATION DIFFERS (SEED 0, FIVE FOLDS).

Dataset	Check-in level	Place level	Δ
Istanbul	**35.35**	29.07	+6.29
AL	**30.77**	29.15	+1.62
AZ	**34.51**	31.93	+2.58
FL	**37.36**	37.13	+0.23
TX	**36.32**	35.33	+0.99
CA	**35.62**	34.74	+0.88

All five folds favor the check-in-level representation at every dataset. The difference is significant under a paired test at every dataset except FL ($p = 0.07$).

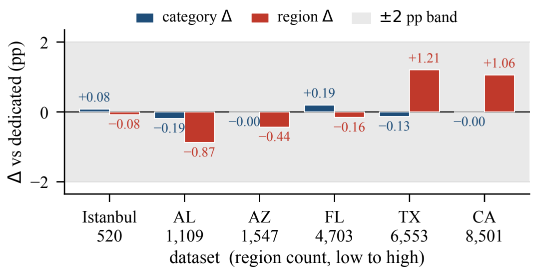

Fig. 3. Signed differences between the joint model and the dedicated single-task models, ordered by region count. The category differences (macro-F1) are small in both directions; on the region axis (Acc@10), the two positive differences at the largest region vocabularies survive a Holm-corrected superiority test, and all six lie inside the band, which is where non-inferiority is established. The ±2-point band is the margin registered for the region axis only.

a tenth of a point of the dedicated models: category $+0.08$ macro-F1 (dedicated 35.34, joint 35.42) and region -0.08 Acc@10 (dedicated 75.16, joint 75.08), the region difference staying within the two-point margin. The comparable quantity is the difference against the dedicated model, not the absolute Acc@10, since region counts differ across datasets (520 mahalle here). The external baselines show the same ordering on both tasks (Table II). What repeats on a different continent and region unit is the bounded picture: the region difference stays within the registered two-point margin and the category difference is equivalent to zero within half a point, with no gain on either task.

VII. DISCUSSION AND LIMITATIONS

A single model serves both tasks in one forward pass, on next region it outperforms the dedicated model at Texas and California, the two datasets with the largest region vocabularies, and at the other four it stays within the two-point margin registered before any result was read. On next category it outperforms the dedicated model at Florida, and the five remaining differences are equivalent to zero within half a point. Throughout, the region output keeps its own private spatial path (Table II; Fig. 3). The practical reading is that one model replaces two while staying inside the margin that the analysis plan registered as acceptable, and does better than two at the two datasets with the largest region vocabularies. The four region results inside that margin are all small deficits rather than ties, the largest of them 0.87 Acc@10 points at Alabama, so the trade is a measured one and not a free substitution.

The design gives the two tasks a shared representation while keeping a private spatial path for the region output. The evidence here does not separate their contributions. Where the trunk was isolated directly, the ablation moved both tasks by small amounts that a screen of that size cannot distinguish from noise, and it was not run at the two datasets that carry the region advantage. The reading this supports is a cautious one: the shared representation appears able to help, and how much it helps is a question for further work rather than one this study settles. What can be stated without that work is narrower, and it is a claim about the model rather than about transfer between its tasks: this design, shared representation and private path together, produces a joint region output above two dedicated models at the two datasets with the largest region vocabularies.

The category picture at Texas and California invites a specific hypothesis. Batch size was searched for the dedicated category model at every dataset and the learning rate at four of them, with both searched for the joint model at the smaller datasets; at Texas and California the joint model runs a configuration transferred from those searches rather than one validated on them. Texas carries the second-largest category deficit of the six (-0.13; Alabama's -0.19 is the largest, and it is a dataset where the joint model was searched), and California's is under a hundredth of a point. A search conducted for the joint model at those two datasets would test whether the transferred configuration is what holds their category output slightly below the dedicated model. Until it is run this remains a hypothesis rather than an explanation, and Alabama is a reminder that a searched dataset can show a deficit of the same size.

A service could treat the model's top-ranked next regions as an anticipatory set: at California, ten regions out of 8,501 contain the true next region 64.54 percent of the time, and at Texas, ten out of 6,553 contain it 66.15 percent of the time. Both figures are the joint model's next-region Acc@10 from Table II, read here as a shortlist hit rate: a system that surfaced ten census tracts would be right about two times in three, over a candidate set three orders of magnitude larger. Where the shortlist misses, the geographic size of the error is the quantity that would matter to such a service, and measuring it requires the per-visit predictions that the evaluation path does not retain, so it is left to future work. This remains motivation, not a measured service result.

Five limits qualify these results. First, the representation is trained once over all places; a per-fold rebuild from training users only changed the results by at most 0.33 Acc@10 and 0.29 macro-F1 across three datasets at one seed, with the qualification on the region half recorded in Section V-B. Second, epoch selection consults the fold that the score is then read on (Section V-B), so every absolute score reported here is optimistic. The comparison between the joint model and the dedicated models is affected far less, for two reasons that we can state rather than assume: the selection rule is the same for both models on the same folds, an epoch chosen on validation and read on that fold, with each model selected on its own validation objective (Section VI-B), and, for the category comparison, the dedicated model receives the wider search: batch size at all six datasets and the learning rate at four, against a joint-model search that covers four of the six and leaves Texas and California with a transferred configuration, while both sides of the region comparison run a fixed configuration (Section V-B states the coverage per knob and dataset). Where the dedicated search is the wider of the two, the residual favors the dedicated model, which makes the reported category difference conservative there. At Florida and California, where the dedicated learning rate was not varied, the two searches are closer in extent and the reading does not apply; Florida is also the one dataset where the joint model outperforms on this task. It does not follow that the bias cancels exactly. Third, we do not build or evaluate a mobility-aware service; it is background motivation, and this paper's claims are the prediction results themselves. Fourth, each visit node in the representation graph draws on the visits that precede it, so a vector encodes the visit together with its own history; the graph does not pass information from a later visit back to an earlier one, in training or at readout, which is what keeps a node from carrying a feature of the target it is used to predict. Fifth, the joint model has more parameters than the two dedicated models together (Section IV-B), so its region result at Texas and California may come from size rather than from sharing. A control separates the two at California: it widens the dedicated region model until its parameter count reaches the joint model's, taken from the training logs, and changes nothing else. At width 352 the dedicated model has $5{,}014{,}942$ parameters, 97.4 percent of the joint model's $5{,}151{,}189$, and scores 0.41 Acc@10 above the joint model at the same random initialization, with all five folds in the same direction and a paired test over the five folds that separates the difference from zero ($p = 0.010$). A wider model, at 174.8 percent of the joint model's budget, scores 0.43 above it ($p = 0.008$, five of five folds); the two are 0.02 apart ($p = 0.40$), so widening past parity adds nothing. Texas has no control of matched size: the only wider model run there has 170.5 percent of the joint model's budget and scores 0.21 above the joint model, which the same test does not separate from zero ($p = 0.12$). At California, the one dataset where the control exists, a dedicated model of the same size scores above the joint model's region result, so the advantage there cannot be attributed to sharing between the tasks; at Texas the question is open. What remains on the region axis is the operational result: one model produces both predictions in one forward pass, at no cost beyond the bounds reported above.

Our representation is a fixed per-visit vector that any downstream model can consume. Moura et al. [5], whose structural analysis of tourist check-ins reads past visits only, name machine learning as a next step; this study moves in that direction.

VIII. Conclusion

We tested whether one model can predict both the category and region of a user's next visit. The central concern was that sharing could help one task while hurting the other. However, the evidence does not attribute the outcome to sharing alone: on the category task the input representation moves the score more than the choice between one model and two at every one of the six datasets, and on the region task a dedicated model of the joint model's size scores above it at California. A standard place-level representation assigns the same vector to every visit to a place, so it cannot distinguish visits made under different contexts. By assigning one vector to each visit, the check-in-level representation can include the visit time, nearby places, and recent visits. The controlled comparison then shows a consistent advantage for that representation on the category task: every fold favors it at every dataset, though the size of the advantage varies from $+0.23$ to $+6.29$ macro-F1 points. Once this information is available, the joint model can share semantic context between the tasks while keeping a private spatial path for region prediction, and read both answers from one saved model.

With both changes in place, the region cost of joint training stays inside the registered two-point margin at every dataset. For region prediction the model outperforms the dedicated model at Texas and California, and at the other four datasets it stays within the two-point margin registered before any result was read. For category prediction it outperforms the dedicated model at Florida; the five remaining differences are not resolved, and every one of them is equivalent to zero within half a point. The two datasets where the region output improves are the two with the largest region vocabularies, which is where an auxiliary signal has the most to add, although region count and corpus size co-vary across these states and the ordering is not monotone within the pair; at California, a dedicated region model of the same size scores above the joint model, so the gain there is not attributed to sharing (Section VII). The joint model also remains above every external baseline in Table II at all six datasets, by at least 3.55 Acc@10 points over the strongest region reference (HMT-GRN, STAN, or ReHDM; the first two on our folds, ReHDM under its own protocol) and by at least 3.06 macro-F1 points over POI-RGNN on category.

Together, these results do not mean that multi-task learning helps automatically. They show that the initial concern about harmful sharing was incomplete: the differences between one model and two are bounded on both axes, and the representation is the larger effect on category. Whether the exchange of information between the tasks contributes anything beyond that is not separated by the evidence here (Section VII). When the representation preserves the context of each visit and the architecture keeps a private spatial path where the tasks differ, one model can predict both what a user will do next and where in a single forward pass instead of two.

REFERENCES

[1] E. Cho, S. A. Myers, and J. Leskovec, “Friendship and mobility: User movement in location-based social networks,” in *Proc. ACM SIGKDD (KDD)*. ACM, 2011, pp. 1082–1090.

[2] C. Song, Z. Qu, N. Blumm, and A.-L. Barabási, “Limits of predictability in human mobility,” *Science*, vol. 327, no. 5968, pp. 1018–1021, 2010.

[3] E. Bastug, M. Bennis, and M. Debbah, “Living on the edge: The role of proactive caching in 5G wireless networks,” *IEEE Commun. Mag.*, vol. 52, no. 8, pp. 82–89, 2014.

[4] C. L. Vielhaus, J. V. S. Busch, P. Geuer, A. Palaios, J. Rischke, D. F. Külzer, V. Latzko, and F. H. P. Fitzek, “Handover predictions as an enabler for anticipatory service adaptations in next-generation cellular networks,” in *Proc. ACM MobiWac*. ACM, 2022, pp. 19–27.

[5] D. L. L. Moura, A. L. L. Aquino, and A. A. F. Loureiro, “On the design of mobility-aware systems: A tourist’s perspective,” in *Proc. IEEE/ACM MSWiM*, 2025, pp. 667–674.

[6] R. Caruana, “Multitask learning,” *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, 1997.

[7] V. H. O. Silva, I. F. Almeida, T. S. Paiva, G. B. Santos, F. A. Silva, and F. T. Sousa, “An investigation into multi-task learning for point-of-interest category classification and next-POI prediction,” in *Anais do CBIC*, 2025.

[8] W. Huang, D. Zhang, G. Mai, X. Guo, and L. Cui, “Learning urban region representations with POIs and hierarchical graph infomax,” *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 134–145, 2023.

[9] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, “Deep graph infomax,” in *Proc. ICLR*, 2019.

[10] W. Wongso, H. Xue, and F. D. Salim, “Massive-steps: Massive semantic trajectories for understanding POI check-ins — dataset and benchmarks,” arXiv:2505.11239, 2025.

[11] Y. Lin, H. Wan, S. Guo, and Y. Lin, “Pre-training context and time aware location embeddings from spatial-temporal trajectories for user next location prediction,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 4241–4248.

[12] J. Feng, Y. Li, C. Zhang, F. Sun, F. Meng, A. Guo, and D. Jin, “DeepMove: Predicting human mobility with attentional recurrent networks,” in *Proc. Web Conf. (WWW)*, 2018, pp. 1459–1468.

[13] Y. Luo, Q. Liu, and Z. Liu, “STAN: Spatio-temporal attention network for next location recommendation,” in *Proc. Web Conf. (WWW)*, 2021, pp. 2177–2185.

[14] S. Yang, J. Liu, and K. Zhao, “GETNext: Trajectory flow map enhanced transformer for next POI recommendation,” in *Proc. ACM SIGIR*, 2022, pp. 1144–1153.

[15] M. Luca, G. Barlacchi, B. Lepri, and L. Pappalardo, “A survey on deep learning for human mobility,” *ACM Comput. Surv.*, vol. 55, no. 1, pp. 7:1–7:44, 2021.

[16] Z. Sun, Y. Lei, L. Zhang, C. Li, Y.-S. Ong, and J. Zhang, “A multi-channel next POI recommendation framework with multi-granularity check-in signals,” *ACM Trans. Inf. Syst.*, vol. 42, no. 1, pp. 1–28, 2024.

[17] N. Lim, B. Hooi, S. Ng, Y. L. Goh, R. Weng, and R. Tan, “Hierarchical multi-task graph recurrent network for next POI recommendation,” in *Proc. ACM SIGIR*, 2022, pp. 1133–1143.

[18] N. Zhu, J. Cao, X. Lu, C. Liu, H. Liu, Y. Li, X. Luo, and H. Xiong, “Predicting a person’s next activity region with a dynamic region-relation-aware graph neural network,” *ACM Trans. Knowl. Discov. Data*, vol. 16, no. 6, pp. 1–23, 2022.

[19] K. Sun and M. Xu, “Knowledge graph tokenization for behavior-aware generative next POI recommendation,” arXiv:2509.12350, 2025.

[20] D. Liao, W. Liu, Y. Zhong, J. Li, and G. Wang, “Predicting activity and location with multi-task context aware recurrent neural network,” in *Proc. IJCAI*, 2018, pp. 3435–3441.

[21] Y. Wang, H. Chen, S. Liu, J. Zhang, L. Wu, X. Cui, and Y. Hu, “Hierarchy aware-based multi-task learning for user location prediction,” *The Journal of Supercomputing*, vol. 81, no. 11, 2025.

[22] J. Ye, Z. Zhu, and H. Cheng, “What’s your next move: User activity prediction in location-based social networks,” in *Proc. SIAM SDM*, 2013, pp. 171–179.

[23] Z. Huang, S. Xu, M. Wang, H. Wu, Y. Xu, and Y. Jin, “Human mobility prediction with causal and spatial-constrained multi-task network,” *EPJ Data Sci.*, vol. 13, no. 1, p. 22, 2024.

[24] F. Yu, L. Cui, W. Guo, X. Lu, Q. Li, and H. Lu, “A category-aware deep model for successive POI recommendation on sparse check-in data,” in *Proc. Web Conf. (WWW)*, 2020, pp. 1264–1274.

[25] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, “Gradient surgery for multi-task learning,” in *Proc. NeurIPS*, vol. 33, 2020.

[26] A. Navon, A. Shamsian, I. Achituve, H. Maron, K. Kawaguchi, G. Chechik, and E. Fetaya, “Multi-task learning as a bargaining game,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 17–23 Jul 2022, pp. 16 428–16 446. [Online]. Available: https://proceedings.mlr.press/v162/navon22a.html

[27] D. Xin, B. Ghorbani, A. Garg, O. Firat, and J. Gilmer, “Do current multi-task optimization methods in deep learning even help?” in *Proc. NeurIPS*, vol. 35, 2022.

[28] T. H. Silva, A. C. Viana, F. Benevenuto, L. Villas, J. Salles, A. A. F. Loureiro, and D. Quercia, “Urban computing leveraging location-based social network data: A survey,” *ACM Comput. Surv.*, vol. 52, no. 1, pp. 17:1–17:39, 2019.

[29] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, “Long-tail learning via logit adjustment,” in *Proc. ICLR*, 2021.

[30] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[31] D. Lakens, “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses,” *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017.

[32] C. G. S. Capanema, G. S. de Oliveira, F. A. Silva, T. R. de Moura Braga Silva, and A. A. F. Loureiro, “Combining recurrent and graph neural networks to predict the next place’s category,” *Ad Hoc Netw.*, vol. 138, p. 103016, 2023.

[33] X. Li, Z. Gu, R. Yao, Y. Zhou, H. Zhu, J. Zhao, and W. Du, “Beyond individual and point: Next POI recommendation via region-aware dynamic hypergraph with dual-level modeling,” in *Proc. IJCAI*, 2025, pp. 3081–3089.